

VINAY KUMAR

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SUMMARY

Final-year B.Tech student in Computer Science (AI & ML) with a CGPA of 8.92, experienced in MERN stack development, cloud deployments, and DevOps fundamentals. Skilled in building scalable web applications and working with modern development tools. Possesses foundational knowledge in Machine Learning and a strong understanding of Data Structures & Algorithms. Passionate about developing efficient, real-world solutions and continuously improving technical expertise.

EDUCATION

VIT BHOPAL UNIVERSITY

Bachelor of Technology, Computer Science & Engineering (AI & ML)

Cumulative GPA: 8.92

Bhopal, MP

Sept 2023 - Present

VIDYA BAL BHAWAN SCHOOL

Intermediate (Class 12th): 86%

Delhi, India

April 2023

MAYO INTERNATIONAL SCHOOL

High School (Class 10th): 85%

Delhi, India

April 2021

PROJECTS

VeggieMap — Hyperlocal Vegetable Vendor Finder

[GitHub](#) | [Live](#)

- **Real-time hyperlocal marketplace:** Engineered a live vendor map using MongoDB 2dsphere geospatial indexing and Socket.io Geo-Room partitioning for scalable location broadcasts.
- **Advanced Frontend:** Deployed as a PWA with Supercluster marker clustering, utilizing Zustand and TanStack Query for optimized state management.

Stack: Next.js 16, TypeScript, Node.js, Express, MongoDB, Socket.io, PM2, Nginx

SMS Gateway — Android-based Twilio Alternative

[GitHub](#) | [Live](#)

- **Twilio Alternative:** Engineered a system to turn Android devices into powerful SMS gateways for sending OTPs and campaigns via cellular plans with zero SaaS fees.
- **Full-Stack Infrastructure:** Developed a real-time web dashboard and a clean REST API, utilizing BullMQ and Redis for instant message queuing and zero-delay delivery.

Stack: React.js, Node.js, Express.js, MongoDB, BullMQ, Redis, Kotlin, Firebase (FCM)

system-ctrl — Server Health Monitoring Dashboard

[GitHub](#) | [Live](#)

- **Real-time Monitoring:** Built a dashboard tracking CPU, Memory, and Network metrics via systeminformation with live in-browser log streaming and authenticated restarts.
- **Infrastructure:** Managed server-side workflows using Nginx and external service pings to ensure maximum system availability and monitoring.

Stack: Next.js, TypeScript, Tailwind CSS v4, Shadcn UI, systeminformation, PM2, Nginx

WebPull AI — AI-Powered Documentation Scraper

[GitHub](#) | [Live](#)

- **Core Engineering:** Engineered a full-stack application using Next.js and Express to scrape live documentation websites and instantly convert them into clean, structured Markdown files.
- **Advanced Integrations:** Integrated Google Gemini API for automated URL discovery and content generation, utilizing Server-Sent Events (SSE) for real-time progress and JSZip for client-side packaging.

Stack: Next.js 14, React, TypeScript, Node.js, Express, Crawlee, Google Gemini API

TECHNICAL SKILLS

Languages:	C++, Python, JavaScript, TypeScript
Web Technologies:	MERN Stack, Next.js, Tailwind CSS, Shadcn UI, Socket.io, REST APIs
Cloud & DevOps:	AWS (EC2), Docker, PM2, Nginx, Linux, Git, CI/CD
Databases:	MongoDB, PostgreSQL, SQL, Firebase, Redis
Machine Learning:	NumPy, Pandas, Scikit-learn, Foundational Models

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services

[Link](#)

Cloud Computing — NPTEL, IIT Madras

ACHIEVEMENTS & ACTIVITIES

- **Coding:** Solved 250+ problems on LeetCode, showcasing strong DSA and problem-solving skills.
- **Leadership:** Active member of the university PR and Outreach team, managing community engagement.
- **Technical Migration:** Led successful project migration to AWS EC2, ensuring 100% service availability.